

Luong Hai Long

Computer Vision / Machine Learning / Network Engineering Intern | Python | PyTorch | OpenCV | MATLAB | Cisco IOS
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Professional Summary

Electronics and Telecommunications Engineering senior with project experience in computer vision, machine learning, digital signal processing, computer networks, and wireless communications. Built Python/PyTorch vision and signal-classification workflows, Cisco Packet Tracer networks, and MATLAB communication-system simulations. Seeking an internship where ML software, networking, and telecommunications meet practical engineering constraints.

Technical Skills

Programming: Python, C, C++, MATLAB; Jupyter/IPYNB, Kaggle, Git, GitHub, Typst.
Computer vision and ML: PyTorch, YOLO/Ultralytics, OpenCV, FastALPR, OCR, object detection, 1D CNN, Random Forest, scikit-learn, NumPy, Streamlit.
Networks: Cisco IOS, Packet Tracer, TCP/IP, IPv4/IPv6, VLAN, inter-VLAN routing, STP/PVST, EtherChannel, OSPFv2, DHCP, NAT, WLAN.
Telecommunications and systems: DSP, BER, AWGN, BPSK/QPSK/FSK/QAM, OFDM, MIMO, fading, FFT, embedded Linux, FPGA/SoC, STM32/ESP32, BLE, UART, I2C, SPI.

Education

VNUHCM - University of Science Sep 2022 – Expected Sep 2026
Bachelor of Engineering in Electronics and Telecommunications Engineering
Ho Chi Minh City
High-Quality Program, English-taught; Student ID: 22207056; GPA: 7.2/10.
• Relevant coursework: Artificial Intelligence (9.1/10), Computer Vision, Object-Oriented Programming (9.0/10 final), Fundamentals of Computer Networks (8.0/10), Network Technologies (8.0/10), Wireless Communications (8.5/10), and Embedded Systems Practice (10/10).

Experience

BTP Holdings Jan 2026 – Apr 2026
R&D Embedded Software Engineer Intern
Ho Chi Minh City
• Implemented and tested STM32/ESP32 firmware support across C/C++ HAL modules, RTOS-oriented tasks, UART/Serial diagnostics, TCP/IP, and BLE communication paths.
• Built a C/Win32 serial diagnostic tool for board bring-up and reviewed USB-C PD behavior and power-board schematics to align firmware tests with hardware constraints.

Robotics & IoT Club, VNUHCM - University of Science Nov 2023 – Jul 2024
Technical Team Member
Ho Chi Minh City
• Assembled and tested robotics and IoT prototypes; isolated sensor wiring, power, and firmware issues during bring-up using C/Lua test routines.

Anh Trung Smartphone Store / Xuan Anh Computer 2023 – 2025
Repair Technician (Part-time)
Ho Chi Minh City
• Diagnosed phone, laptop, and PC faults involving power, display, storage, thermal, connectors, software, and peripherals; completed functional tests before handover.

Selected Projects

Vietnamese Automatic License Plate Recognition with YOLO and OCR Jan 2026 – Apr 2026
Team of seven | Python, PyTorch, YOLO, OpenCV, FastALPR, OCR, FFmpeg, Tkinter
• Co-developed image/video desktop inference and a bounded PlateGate LAN whitelist demo; maintained repository structure, Git LFS assets, Kaggle/IPYNB training evidence, and release packaging.
• Evaluated the detector at 99.45% mAP50, 99.45% precision, and 99.37% recall on the project validation set.

Automatic Modulation Classification with DSP Features and 1D CNN Aug 2026
Python, PyTorch, scikit-learn, Random Forest, Streamlit, I/Q signal processing
• Classified six simulated modulation formats using 18 DSP features and an 81,030-parameter 1D CNN; validation-selected fusion reached 73.48% accuracy and 73.40% macro-F1 on a fixed 1,320-frame test split.

Biomedical Image Processing Laboratory Aug 2026
Python, Jupyter, CT imaging, spatial filtering, FFT, interpolation, SSIM
• Implemented grayscale conversion, ROI statistics, explicit 5 x 5 convolution, radix-2 2D FFT/IDFT, Butterworth sharpening, and bilinear/bicubic resampling on course CT samples; documented an eleven-slice 512 x 512 data record and bicubic SSIM of 0.8535.

Network Technology Laboratory 2026
Cisco IOS, Packet Tracer, IPv4/IPv6, VLAN, STP, EtherChannel, OSPF, DHCP, NAT, WLAN
• Configured and verified six laboratories through routing tables, VLAN/trunk state, OSPF adjacencies, DHCP addressing, NAT translation, wireless access, and end-to-end reachability; compiled reproducible Typst reports.

Wireless Communications MATLAB Practice Sep 2025 – Dec 2025
MATLAB, BER Monte Carlo, Rayleigh/Rician fading, equalization, OFDM, MIMO
• Analyzed path loss, fading, channel estimation, OFDM, 2 x 2 MIMO with SVD precoding, and diversity tradeoffs across 23 MATLAB scripts using BER, outage, and capacity plots.